

MIAE 221 · Rapid Review Sheet · Part 9

Diffusion in Solids

1. Fundamental Principles & Atomic Mechanisms

- **Diffusion Definition:** Mass transport through atomic motion driven by thermal energy and chemical potential / concentration gradients.
- **Self-Diffusion:** Atomic motion in pure elements with no net compositional change; tracked via radioactive isotopes.
- **Interdiffusion (Diffusion Couple):** Net mass transport between dissimilar metals (e.g. Cu-Ni) smoothing initial step-function profiles into solid solutions.
- **Prerequisites for Atomic Jump:** (1) Adjacent empty site (vacancy or interstitial void); (2) Thermal vibrational energy $E \geq Q$ to overcome the barrier.
- **Vacancy Diffusion:** Host/substitutional atoms jump into neighboring vacancies. $\vec{J}_{\text{atoms}} = -\vec{J}_{\text{vacancies}}$. Requires vacancy formation + migration: $Q_d = Q_v + Q_m$. Slow rate.
- **Interstitial Diffusion:** Small solutes (C, H, N, O) jump between interstitial voids. No vacancies needed; $Q_d = Q_m$. Much faster ($10^2 - 10^5 \times$) than vacancy diffusion.

2. Steady-State Diffusion & Fick's First Law

- **Diffusion Flux (J):** Mass M (or atoms) passing perpendicular to area A per unit time t :

$$J = \frac{M}{At} = \frac{1}{A} \frac{dM}{dt} \quad \left[\frac{\text{kg}}{\text{m}^2 \cdot \text{s}} \text{ or } \frac{\text{atoms}}{\text{m}^2 \cdot \text{s}} \right]$$

- **Steady-State Definition:** Concentration profile does not vary with time: $\frac{\partial C}{\partial t} = 0 \implies J = \text{constant}$.
- **Fick's 1st Law:** Flux is proportional to the concentration gradient:

$$J = -D \frac{dC}{dx} \approx -D \left(\frac{C_B - C_A}{x_B - x_A} \right)$$

- D : Diffusion coefficient / diffusivity (m^2/s).
- Negative sign: Mass spontaneously flows *down* the concentration gradient.
- For flat membrane at steady state: $\frac{dC}{dx} = \text{constant}$ (linear profile).

3. Non-Steady State Diffusion & Fick's Second Law

- **Non-Steady State:** Concentration changes with time and location: $\frac{\partial C}{\partial t} \neq 0$.
- **Fick's 2nd Law (Constant D):**

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

- **Semi-Infinite Solid Solution** ($l > 10\sqrt{Dt}$, uniform initial C_0 , constant surface C_s at $x = 0$):

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \iff \frac{C_s - C_x}{C_s - C_0} = \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- **Gaussian Error Function:** $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$ with $\text{erf}(0) = 0$, $\text{erf}(\infty) = 1$.
- **Linear Interpolation Formula:**

$$z = z_1 + \left(\frac{\text{erf}(z) - \text{erf}(z_1)}{\text{erf}(z_2) - \text{erf}(z_1)} \right) (z_2 - z_1)$$

- **Quick Reference Table:**

z	$\text{erf}(z)$	z	$\text{erf}(z)$	z	$\text{erf}(z)$
0.20	0.2227	0.45	0.4755	0.80	0.7421
0.30	0.3286	0.50	0.5205	0.90	0.7969
0.40	0.4284	0.60	0.6039	1.00	0.8427

4. Constant-Concentration Scaling Laws

- If a specific target concentration C_x is specified, then $\frac{C_x - C_0}{C_s - C_0} = \text{const} \implies \frac{x}{2\sqrt{Dt}} = \text{const}$:

$$\frac{x^2}{Dt} = \text{constant} \iff \frac{x_1^2}{D_1 t_1} = \frac{x_2^2}{D_2 t_2}$$

- Constant Temperature** ($D_1 = D_2$): $\frac{x_1^2}{t_1} = \frac{x_2^2}{t_2} \implies \frac{x_1}{\sqrt{t_1}} = \frac{x_2}{\sqrt{t_2}}$.
 - To double case depth ($x_2 = 2x_1$), required time **quadruples** ($t_2 = 4t_1$).
- Constant Depth** ($x_1 = x_2$): $D_1 t_1 = D_2 t_2 \implies t_2 = t_1 \left(\frac{D_1}{D_2} \right)$.

5. Temperature Dependence & Activation Energy

- Arrhenius Equation:** $D = D_0 \exp\left(-\frac{Q_d}{RT}\right) = D_0 \exp\left(-\frac{Q_d}{k_B T}\right)$
 - D_0 : Pre-exponential frequency factor (m^2/s).
 - Q_d : Activation energy (J/mol , kJ/mol , or eV/atom).
 - $R = 8.314 \text{ J/mol} \cdot \text{K}$; $k_B = 8.62 \times 10^{-5} \text{ eV/atom} \cdot \text{K}$; T strictly in Kelvin!
- Arrhenius Linearization:**
 - $\ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T} \right) \implies \text{Slope} = -\frac{Q_d}{R}, \quad \text{Intercept} = \ln D_0$
 - $\log_{10} D = \log_{10} D_0 - \frac{Q_d}{2.303R} \left(\frac{1}{T} \right) \implies \text{Slope} = -\frac{Q_d}{2.303R}, \quad \text{Intercept} = \log_{10} D_0$
- Diffusion Pathway Rates:**

$$D_{\text{surface}} > D_{\text{grain boundary}} > D_{\text{dislocation}} > D_{\text{lattice (bulk)}}$$

$$Q_{\text{surface}} < Q_{\text{grain boundary}} < Q_{\text{dislocation}} < Q_{\text{lattice (bulk)}}$$

6. Quick Exam Rules of Thumb & Pitfalls

- Open vs Close-Packed:** Diffusion is **faster in open structures** (BCC APF = 0.68 > FCC APF = 0.74) due to larger interstitial voids and lower Q_d .
- Melting Temperature:** Diffusion is **faster in lower- T_m materials** due to weaker bonds and lower Q_d .
- Bond Character:** Diffusion is **faster in secondary bonded solids** than strongly directional covalent solids.
- Units Check:** Convert $T \rightarrow \text{K}$, $t \rightarrow \text{s}$, $x \rightarrow \text{m}$, and ensure Q_d (J vs kJ) matches R .